

DEPARTMENT OF TRAINING AND PLACEMENT

SQL PROBLEMS

1. PRACTICAL QUESTION – MOVIE DATABASE

You are developing a **Movie Database** for an online streaming platform.

A. Table Creation

Create a table named **Movies** with the following attributes and data types:

Attribute Name	Description	Data Type
movie_id	Unique ID for each movie	INT
movie_name	Name of the movie	VARCHAR(100)
genre	Movie genre	VARCHAR(30)
director	Director of the movie	VARCHAR(50)
release_year	Year the movie was released	INT
rating	Movie rating	DECIMAL(3,1)
budget	Movie budget	INT
language	Language of the movie	VARCHAR(30)

B. Alter Table

Modify the **Movies** table by:

- **Adding a new column** named **duration**
- The duration column should store the movie length in minutes
- Data type of duration should be **INT**

C. Rename Table

Rename the table **Movies** to **Movie_Details**.

D. Insert Data

Insert **at least 10 records** into the **Movie_Details** table using the **sample data given**.

movie_id	movie_name	genre	director	release_year	rating	budget	language	duration
1	Inception	Sci-Fi	Christopher Nolan	2010	8.8	160000000	English	148
2	Interstellar	Sci-Fi	Christopher Nolan	2014	8.6	165000000	English	169
3	Avengers	Action	Joss Whedon	2012	8.0	220000000	English	143
4	Bahubali	Action	Rajamouli	2015	8.1	180000000	Telugu	159
5	RRR	Action	Rajamouli	2022	8.5	200000000	Telugu	182
6	Dangal	Drama	Nitesh Tiwari	2016	8.4	70000000	Hindi	161
7	PK	Comedy	Rajkumar Hirani	2014	NULL	85000000	Hindi	153
8	Titanic	Romance	James Cameron	1997	7.8	200000000	English	195
9	Joker	Drama	Todd Phillips	2019	8.2	55000000	NULL	122
10	Tenet	Sci-Fi	Christopher Nolan	2020	7.5	205000000	English	150

E. Update Data

Update the rating of **movie_id = 5** to 9.5.

F. Delete Data

Delete all movies that were released **before the year 2010**.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display all movie records
2. Display all **distinct genres**
3. Display movies with rating **greater than 7**
4. Display movies where genre is **Action** and release year is **after 2015**
5. Display movies ordered by rating in **descending order**
6. Display the **top 5 highest-rated movies**
7. Display movies where the **rating is NULL**
8. Display movie name, genre, and rating where rating is greater than 6, ordered by rating (descending), showing only **first 3 records**

2. PRACTICAL QUESTION – FLIGHT RESERVATION DATABASE

You are developing a **Flight Reservation Database** for an airline booking system.

A. Table Creation

Create a table named **Flights** with the following attributes and data types:

Attribute Name	Description	Data Type
flight_id	Unique ID for each flight	INT
flight_name	Name of the flight	VARCHAR(100)
source	Departure city	VARCHAR(40)
destination	Arrival city	VARCHAR(40)
airline	Airline company name	VARCHAR(50)
travel_year	Year of travel	INT
ticket_price	Ticket cost	INT
seat_type	Type of seat	VARCHAR(30)

B. Alter Table

Modify the **Flights** table by:

- Adding a new column named **available_seats**
- This column should store the number of seats currently available
- Data type should be **INT**

C. Rename Table

Rename the table **Flights** to **Flight_Details**.

D. Insert Data

Insert **at least 10 records** into the **Flight_Details** table using the sample data given below.

flight_id	flight_name	source	destination	airline	travel_year	ticket_price	seat_type	available_seats
1	AI-202	Delhi	Mumbai	Air India	2023	6500	Economy	45
2	SG-118	Hyderabad	Chennai	SpiceJet	2024	4800	Economy	30
3	6E-445	Bangalore	Delhi	IndiGo	2023	7200	Business	12
4	VJ-900	Kochi	Dubai	Vistara	2025	18000	Business	8
5	AK-77	Mumbai	Singapore	AirAsia	2024	15000	Economy	20
6	BA-12	Delhi	London	British Airways	2025	55000	First	5
7	EK-501	Chennai	Dubai	Emirates	2023	22000	Business	NULL
8	QR-36	Doha	New York	Qatar Airways	2024	60000	First	3
9	AI-404	Pune	Jaipur	Air India	2022	3900	Economy	60
10	SG-909	Goa	Delhi	SpiceJet	2023	5200	NULL	25

E. Update Data

Increase the **available_seats** by 10
for all flights where the **seat_type** is 'Economy'
and the **travel_year** is 2024.

F. Delete Data

Delete all flight records
where the **ticket_price** is greater than 50,000
and the **seat_type** is 'First'.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display only the **flight_name, source, and destination** of all flights
2. Display all **unique airline names** operating flights
3. Display flights that **do not travel from Delhi**
4. Display flights whose **destination is Dubai or London**
5. Display flights where **available_seats is less than 20**
6. Display flights scheduled in the **year 2025**, sorted by **ticket_price (ascending)**
7. Display flights where **seat_type is missing (NULL)**
8. Display the **cheapest 3 flights** traveling **from Mumbai**, showing flight name and ticket price

3. PRACTICAL QUESTION – LIBRARY MEMBERSHIP DATABASE

You are designing a **Library Membership Database** for a public library system.

A. Table Creation

Create a table named **Members** with the following attributes and data types:

Attribute Name	Description	Data Type
member_id	Unique ID for each member	INT
member_name	Name of the member	VARCHAR(100)
membership_type	Type of membership	VARCHAR(30)
city	City of the member	VARCHAR(40)
join_year	Year the member joined	INT
books_issued	Number of books currently issued	INT
fine_amount	Pending fine amount	DECIMAL(6,2)
contact_number	Contact number	VARCHAR(15)

B. Alter Table

Modify the **Members** table by:

- Adding a new column named **email**
- The email column should store the member's email address
- Data type should be **VARCHAR(60)**

C. Rename Table

Rename the table **Members** to **Member_Details**.

D. Insert Data

Insert **at least 10 records** into the **Member_Details** table using the sample data given below.

member_id	member_name	membership_type	city	join_year	books_issued	fine_amount	contact_number	email
1	Ravi Kumar	Gold	Hyderabad	2018	3	150.00	9876543210	ravi@gmail.com
2	Anjali Sharma	Silver	Delhi	2020	1	0.00	9123456780	anjali@gmail.com
3	Suresh Rao	Gold	Bangalore	2019	5	300.00	9988776655	suresh@gmail.com
4	Meena Patel	Platinum	Ahmedabad	2022	2	0.00	9090909090	meena@gmail.com
5	Kiran Das	Silver	Kolkata	2017	0	50.00	9345678123	kiran@gmail.com
6	Pooja Nair	Gold	Kochi	2021	4	200.00	9456123789	pooja@gmail.com
7	Arjun Singh	Platinum	Jaipur	2023	1	NULL	9001122334	arjun@gmail.com
8	Neha Verma	Silver	Pune	2016	6	450.00	9887766554	NULL
9	Ramesh Iyer	Gold	Chennai	2020	2	0.00	9765432109	ramesh@gmail.com
10	Kavya Reddy	Platinum	Hyderabad	2024	1	0.00	9556677889	kavya@gmail.com

E. Update Data

Update the **membership_type** to 'Platinum'
for all members who have **issued more than 4 books**
and have a **fine_amount greater than 200**.

F. Delete Data

Delete all member records
where the **books_issued is 0**
and the **fine_amount is less than 100**.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display the **member_name and city** of members who joined **after 2019**
2. Display all **different membership types** along with the **count of members** in each
3. Display members whose **fine_amount is greater than the average fine amount**
4. Display members from **Hyderabad or Chennai** who have **issued at least 2 books**
5. Display member details sorted by **books_issued (descending)** and **join_year (ascending)**
6. Display the **oldest 3 members** based on **join_year**
7. Display members whose **email address is not available**
8. Display the **top 2 cities** having the **highest number of members**

4. PRACTICAL QUESTION – PRODUCT INVENTORY DATABASE

You are designing a **Product Inventory Database** for a retail store.

A. Table Creation

Create a table named **Products** with the following attributes and data types:

Attribute Name	Description	Data Type
product_id	Unique ID for each product	INT
product_name	Name of the product	VARCHAR(100)
category	Product category	VARCHAR(40)
supplier	Supplier name	VARCHAR(50)
manufacture_year	Year of manufacture	INT
price	Product price	DECIMAL(8,2)
stock_quantity	Available stock	INT
brand	Product brand	VARCHAR(40)

B. Alter Table

Modify the **Products** table by:

- Adding a new column named **warranty_years**
- This column should store warranty period in years
- Data type should be **INT**

C. Rename Table

Rename the table **Products** to **Store_Inventory**.

(Note: The renamed table should represent all products available in the store.)

D. Insert Data

Insert **at least 10 records** into the **Store_Inventory** table using the sample data given below.

product_id	product_name	category	supplier	manufacture_year	price	stock_quantity	brand	warranty_years
1	Laptop Pro	Electronics	TechSupply	2022	85000.00	25	Dell	3
2	Smart TV	Electronics	VisionWorld	2021	62000.00	18	Samsung	2
3	Office Chair	Furniture	HomePlus	2020	7500.00	40	Nilkamal	5
4	Mixer Grinder	Appliances	KitchenCare	2019	4200.00	30	Philips	2
5	Smartphone X	Electronics	MobileHub	2023	72000.00	15	Apple	1
6	Washing Machine	Appliances	HomeTech	2021	38000.00	12	LG	2
7	Bookshelf	Furniture	WoodCraft	2018	5600.00	8	IKEA	NULL
8	Air Conditioner	Appliances	CoolAir	2022	54000.00	10	Voltas	3
9	Headphones	Electronics	SoundZone	2020	3500.00	50	Sony	1
10	Coffee Maker	Appliances	BrewMaster	2019	6800.00	0	Morphy	NULL

E. Update Data

Increase the **warranty_years** by 1
for all products that belong to the **Electronics** category
and were **manufactured after 2021**.

F. Delete Data

Delete all products
where the **stock_quantity** is 0
and the **warranty_years** is NULL.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display the **product_name** and **brand** of products priced between **5000** and **60000**
2. Display the **total number of products** available in each category
3. Display products supplied by **'TechSupply'** or **'MobileHub'**
4. Display products whose **brand name starts with the letter 'S'**
5. Display products with **warranty_years greater than 2**, sorted by **price (ascending)**
6. Display the **most expensive product** in the inventory
7. Display products for which **warranty information is missing**
8. Display the **top 3 categories** having the **highest total stock quantity**

5. PRACTICAL QUESTION – HOSPITAL APPOINTMENT DATABASE

You are developing a **Hospital Appointment Database** for managing patient appointments.

A. Table Creation

Create a table named **Appointments** with the following attributes and data types:

Attribute Name	Description	Data Type
appointment_id	Unique ID for each appointment	INT
patient_name	Name of the patient	VARCHAR(100)
department	Hospital department	VARCHAR(40)
doctor_name	Doctor assigned	VARCHAR(50)
appointment_year	Year of appointment	INT
consultation_fee	Consultation fee	DECIMAL(7,2)
appointment_status	Status of appointment	VARCHAR(20)
contact_number	Patient contact number	VARCHAR(15)

B. Alter Table

Modify the **Appointments** table by:

- Adding a new column named **follow_up_required**
- This column should store whether a follow-up is needed (Yes / No)
- Data type should be **VARCHAR(5)**

C. Rename Table

Rename the table **Appointments** to **Active_Appointments**.

D. Insert Data

Insert **at least 10 records** into the **Active_Appointments** table using the sample data given below.

appointment_id	patient_name	department	doctor_name	appointment_year	consultation_fee	appointment_status	contact_number	follow_up_required
1	Ramesh Kumar	Cardiology	Dr. Mehta	2023	1200.00	Completed	9876543210	No
2	Anita Rao	Neurology	Dr. Sen	2024	1500.00	Scheduled	9123456789	Yes
3	Suresh Patel	Orthopedics	Dr. Shah	2023	900.00	Completed	9988776655	No
4	Pooja Nair	Dermatology	Dr. Iyer	2022	700.00	Cancelled	9090909090	No
5	Karthik Reddy	Cardiology	Dr. Mehta	2024	1300.00	Scheduled	9345678123	Yes
6	Meena Joshi	Pediatrics	Dr. Kulkarni	2023	800.00	Completed	9456123789	No
7	Arjun Singh	Neurology	Dr. Sen	2024	1600.00	Scheduled	9001122334	Yes
8	Neha Verma	Orthopedics	Dr. Shah	2022	950.00	Completed	9887766554	No
9	Vinay Das	Dermatology	Dr. Iyer	2023	650.00	Cancelled	9765432109	NULL
10	Kavita Malhotra	Pediatrics	Dr. Kulkarni	2024	850.00	Scheduled	9556677889	Yes

E. Update Data

Change the **appointment_status** to 'Completed'
for all appointments where:

- **follow_up_required** = 'No'
- and **appointment_year** = 2024

F. Delete Data

Delete all appointment records
where the **appointment_status** is 'Cancelled'
and the **consultation_fee** is less than 800.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display the **patient_name** and **department** for appointments handled by **Dr. Mehta**
2. Display the **number of appointments** scheduled in each department
3. Display appointments where the **consultation_fee** is exactly 900 or 1200
4. Display appointments that **do not require follow-up**
5. Display appointments scheduled in **2024**, sorted by **department (alphabetical order)**
6. Display the **department with the highest average consultation fee**
7. Display appointments where **follow_up_required** is NULL
8. Display the **first 2 scheduled appointments** based on **appointment_id**

6. PRACTICAL QUESTION – EMPLOYEE PAYROLL DATABASE

You are developing an **Employee Payroll Database** for an organization.

A. Table Creation

Create a table named **Employees** with the following attributes and data types:

Attribute Name	Description	Data Type
employee_id	Unique ID for each employee	INT
employee_name	Name of the employee	VARCHAR(100)
department	Department of the employee	VARCHAR(40)
designation	Job role	VARCHAR(40)
joining_year	Year of joining	INT
salary	Monthly salary	DECIMAL(8,2)
employment_type	Employment type	VARCHAR(20)
city	City of work location	VARCHAR(40)

B. Alter Table

Modify the **Employees** table by:

- Adding a new column named **bonus**
- This column should store the annual bonus amount
- Data type should be **DECIMAL(7,2)**

C. Rename Table

Rename the table **Employees** to **Payroll_Records**.

D. Insert Data

Insert **at least 10 records** into the **Payroll_Records** table using the sample data given below.

employee_id	employee_name	department	designation	joining_year	salary	employment_type	city	bonus
1	Aman Verma	IT	Developer	2019	65000.00	Full-Time	Bangalore	5000.00
2	Sneha Rao	HR	Manager	2017	78000.00	Full-Time	Mumbai	8000.00
3	Rakesh Singh	Finance	Analyst	2021	54000.00	Contract	Delhi	3000.00
4	Neha Patel	IT	Tester	2020	48000.00	Full-Time	Ahmedabad	NULL
5	Kunal Mehta	Sales	Executive	2018	45000.00	Full-Time	Pune	4000.00
6	Priya Nair	IT	Developer	2022	60000.00	Contract	Kochi	2000.00
7	Arjun Das	Finance	Manager	2016	82000.00	Full-Time	Kolkata	10000.00
8	Kavita Joshi	HR	Recruiter	2023	42000.00	Intern	Jaipur	NULL
9	Rohit Malhotra	Sales	Manager	2019	70000.00	Full-Time	Delhi	6000.00
10	Meera Iyer	IT	Architect	2015	90000.00	Full-Time	Chennai	12000.00

E. Update Data

Increase the **bonus by 2000**
for employees who are **Full-Time**
and whose **salary is BETWEEN 60000 and 80000**.

F. Delete Data

Delete all employee records
where the **employment_type** is 'Intern' OR 'Contract'
and the **salary is less than 50000**.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display employee names and salaries of employees working in **IT or Finance** departments
2. Display employees whose **salary is BETWEEN 50000 and 75000**
3. Display employees who are **not working in Bangalore**
4. Display employees whose **city is IN (Delhi, Mumbai, Chennai)**
5. Display employees who joined **before 2018 AND** belong to **Sales department**
6. Display employees whose **bonus is NOT NULL**
7. Display employees whose **designation is NOT IN ('Manager', 'Architect')**
8. Display employees whose **salary is greater than the average salary**
9. Display the **total number of employees** in each department
10. Display departments where the **average salary is greater than 65000**
11. Display the **highest-paid employee** in the organization
12. Display employee name, department, and salary of the **top 3 salaries**, sorted by salary (descending)

7. PRACTICAL QUESTION – VEHICLE SERVICE DATABASE

You are developing a **Vehicle Service Database** for an automobile service center.

A. Table Creation

Create a table named **Vehicles** with the following attributes and data types:

Attribute Name	Description	Data Type
vehicle_id	Unique ID for each vehicle	INT
owner_name	Name of the vehicle owner	VARCHAR(100)
vehicle_type	Type of vehicle	VARCHAR(30)
brand	Vehicle brand	VARCHAR(40)
service_year	Year of last service	INT
service_cost	Cost of service	DECIMAL(7,2)
fuel_type	Type of fuel	VARCHAR(20)
city	City of service	VARCHAR(40)

B. Alter Table

Modify the **Vehicles** table by:

- Adding a new column named **service_status**
- This column should store the service status (Completed / Pending)
- Data type should be **VARCHAR(15)**

C. Rename Table

Rename the table **Vehicles** to **Service_Registry**.

D. Insert Data

Insert **at least 10 records** into the Service_Registry table using the sample data given below.

vehicle_id	owner_name	vehicle_type	brand	service_year	service_cost	fuel_type	city	service_status
1	Rajesh Kumar	Car	Hyundai	2023	6500.00	Petrol	Bangalore	Completed
2	Anita Sharma	Bike	Honda	2024	1800.00	Petrol	Delhi	Pending
3	Mohan Das	Car	Toyota	2022	7200.00	Diesel	Kochi	Completed
4	Priya Verma	Scooter	TVS	2023	1500.00	Petrol	Jaipur	Completed
5	Suresh Rao	Car	Tata	2021	8200.00	Diesel	Hyderabad	Pending
6	Kavya Nair	Bike	Yamaha	2024	2200.00	Petrol	Chennai	Completed
7	Arjun Singh	Car	Mahindra	2020	9000.00	Diesel	Pune	Completed
8	Neha Patel	Scooter	Suzuki	2022	1600.00	Petrol	Ahmedabad	NULL
9	Vinod Iyer	Car	Kia	2023	6800.00	Petrol	Coimbatore	Completed
10	Pooja Mehta	Bike	Royal Enfield	2021	3500.00	Petrol	Mumbai	Pending

E. Update Data

Update the **service_status** to 'Completed'
for all vehicles whose **service_year** is 2024
and **vehicle_type** is 'Bike'.

F. Delete Data

Delete all vehicle records
where the **service_cost** is less than 2000
and the **service_status** is 'Pending'.

QUERY OPERATIONS

Answer the following using SQL queries:

1. Display the **owner_name** and **brand** of all vehicles
2. Display all **distinct vehicle types**
3. Display vehicles whose **service_cost** is **BETWEEN 5000 and 8000**
4. Display vehicles where **fuel_type** is **IN ('Petrol', 'Diesel')**
5. Display vehicles serviced in **2023 AND located in Bangalore**
6. Display vehicles serviced in **2021 OR 2022**
7. Display vehicles that are **NOT serviced in Chennai**
8. Display vehicles where the **service_status** is **NULL**
9. Display vehicles whose **brand** is **NOT IN ('Honda', 'TVS')**
10. Display vehicles ordered by **service_cost** in **ascending order**
11. Display the **first 4 vehicles** based on **service_year**
12. Display **owner_name, vehicle_type, and city** for vehicles whose **service_cost** is **greater than 6000**, ordered by **service_cost** (descending)

8. PRACTICAL QUESTION – HOTEL ROOM BOOKING DATABASE

You are developing a **Hotel Room Booking System** for a chain of luxury hotels.

A. Table Creation

Create a table named **Room_Bookings** with the following attributes:

Column Name	Description	Data Type
booking_id	Unique booking ID	INT
guest_name	Name of the guest	VARCHAR(100)
room_type	Type of room	VARCHAR(30)
hotel_name	Name of the hotel	VARCHAR(100)
booking_year	Year of booking	INT
room_price	Price per stay	DECIMAL(8,2)
payment_mode	Mode of payment	VARCHAR(30)
city	Hotel city	VARCHAR(40)

B. Alter Table

Modify the table by:

- Adding a new column named **checkin_status**
- This column stores the check-in state of the guest
- Data type should be VARCHAR(20)

C. Rename Table

Rename **Room_Bookings** to **Guest_Stay_Records**.

D. Insert Data

Insert at least 10 records into **Guest_Stay_Records** using the sample data below:

booking_id	guest_name	room_type	hotel_name	booking_year	room_price	payment_mode	city	checkin_status
1	Arun Kumar	Deluxe	Taj Palace	2023	8500.00	Card	Delhi	Checked-In
2	Meera Joshi	Suite	Oberoi Grand	2024	15000.00	UPI	Kolkata	NULL
3	Ramesh Iyer	Standard	ITC Grand	2022	4500.00	Cash	Chennai	Checked-Out
4	Pooja Shah	Deluxe	Taj Palace	2021	7000.00	Card	Mumbai	Checked-Out
5	Kiran Patel	Suite	Leela Palace	2024	18000.00	Card	Bangalore	Checked-In
6	Sneha Roy	Standard	Oberoi Grand	2023	5200.00	UPI	Kolkata	NULL
7	Ajay Verma	Deluxe	ITC Grand	2022	6800.00	Cash	Chennai	Checked-In
8	Nandini Rao	Suite	Leela Palace	2024	20000.00	Card	Hyderabad	NULL
9	Mohit Jain	Standard	Taj Palace	2020	4000.00	Cash	Delhi	Checked-Out
10	Kavya N	Deluxe	ITC Grand	2023	7500.00	UPI	Bangalore	Checked-In

E. Update Data

Update **checkin_status** to 'Checked-Out'
for guests who:

- Paid using **Cash**
- AND booked in the year **2022**

F. Delete Data

Delete booking records where:

- **room_price** < 5000
- AND **booking_year** is before 2022

QUERY OPERATIONS

1. Display bookings where **booking_year** BETWEEN 2022 AND 2024 AND **room_price** NOT BETWEEN 6000 AND 12000
2. Display records where **room_type** IN ('Suite', 'Deluxe') AND **payment_mode** = 'Card'
3. Display guests staying in **Delhi** OR **Bangalore** AND NOT booked in 2020
4. Display bookings where **checkin_status** IS NULL OR **room_price** > 15000
5. Display bookings where **city** NOT IN ('Mumbai', 'Chennai') AND **room_type** = 'Standard'
6. Display guests who booked BETWEEN 2021 AND 2023 AND **payment_mode** NOT IN ('Cash')
7. Display records where **booking_year** = 2024 OR **room_price** BETWEEN 7000 AND 9000
8. Display bookings where **hotel_name** NOT IN ('Taj Palace', 'ITC Grand') AND **checkin_status** = 'Checked-In'
9. Display guests from **Kolkata** OR **Hyderabad** AND **checkin_status** IS NULL
10. Display all records ordered by **room_price** DESC, **booking_year** ASC
11. Display the latest 4 bookings based on **booking_year**
12. Display **guest_name**, **hotel_name**, **room_price** where **room_price** > 8000 AND **booking_year** NOT IN (2020, 2021) ordered by **room_price** DESC

9. PRACTICAL QUESTION – RESTAURANT TABLE RESERVATION DATABASE

You are developing a **Restaurant Table Reservation System** for a chain of premium restaurants.

A. Table Creation

Create a table named **Reservations** with the following attributes:

Column Name	Description	Data Type
reservation_id	Unique reservation ID	INT
customer_name	Name of the customer	VARCHAR(100)
restaurant_name	Name of the restaurant	VARCHAR(100)
city	Restaurant city	VARCHAR(40)
reservation_date	Date of reservation	DATE
reservation_time	Time of reservation	TIME
number_of_guests	Number of guests	INT
is_confirmed	Reservation confirmation status	BOOLEAN
booking_datetime	Date and time of booking	DATETIME

B. Alter Table

Modify the table by:

- Adding a new column named **special_request**
- This column stores special instructions from customers
- Data type should be VARCHAR(100)

C. Rename Table

Rename the table **Reservations** to **Dining_Reservations**.

D. Insert Data

Insert **at least 10 records** into **Dining_Reservations** using the sample data below:

reservation_id	customer_name	restaurant_name	city	reservation_date	reservation_time	number_of_guests	is_confirmed	booking_datetime	special_request
1	Rahul Mehta	Barbeque Nation	Bangalore	2024-06-15	19:30:00	4	TRUE	2024-06-10 10:15:00	Window seat
2	Ananya Rao	Absolute Barbecues	Hyderabad	2024-06-18	20:00:00	6	TRUE	2024-06-12 14:40:00	NULL
3	Karthik N	Paradise Biryani	Hyderabad	2024-06-20	13:00:00	3	FALSE	2024-06-18 09:30:00	Birthday
4	Pooja Shah	Flechazo	Mumbai	2024-05-10	21:00:00	2	TRUE	2024-05-01 18:00:00	Anniversary
5	Aman Verma	Mainland China	Delhi	2024-07-01	20:30:00	5	TRUE	2024-06-25 11:20:00	NULL
6	Sneha Roy	Barbeque Nation	Kolkata	2024-06-05	19:00:00	4	FALSE	2024-06-03 08:10:00	Vegan food

reservation_id	customer_name	restaurant_name	city	reservation_date	reservation_time	number_of_guests	is_confirmed	booking_datetime	special_request
7	Rohan Patel	Absolute Barbecues	Ahmedabad	2024-07-03	18:30:00	8	TRUE	2024-06-28 16:45:00	Corner table
8	Neha Gupta	Flechazo	Pune	2024-06-25	20:00:00	2	TRUE	2024-06-20 13:00:00	NULL
9	Mohit Jain	Paradise Biryani	Chennai	2024-04-15	14:00:00	3	FALSE	2024-04-12 12:30:00	NULL
10	Kavya N	Mainland China	Bangalore	2024-07-05	21:30:00	6	TRUE	2024-07-01 09:00:00	Birthday

E. Update Data

Update **is_confirmed** to **TRUE**
for reservations that:

- Have **reservation_date** in **July 2024**
- AND **number_of_guests** is greater than 4

F. Delete Data

Delete reservation records where:

- **is_confirmed = FALSE**
- AND **reservation_date** is before **2024-05-01**

QUERY OPERATIONS

1. Display reservations where **reservation_date** BETWEEN '2024-06-01' AND '2024-06-30' AND **number_of_guests** NOT BETWEEN 3 AND 5
2. Display records where **restaurant_name** IN ('Barbeque Nation', 'Flechazo') AND **is_confirmed** = TRUE
3. Display customers from **Hyderabad** OR **Bangalore** AND NOT having reservations in **April 2024**
4. Display reservations where **special_request** IS NULL OR **number_of_guests** > 6
5. Display reservations where **city** NOT IN ('Mumbai', 'Delhi') AND **is_confirmed** = TRUE
6. Display customers who booked BETWEEN '2024-06-01' AND '2024-07-01' AND **booking_datetime** NOT BETWEEN '2024-06-01 00:00:00' AND '2024-06-15 23:59:59'
7. Display records where **reservation_time** BETWEEN '19:00:00' AND '21:00:00' OR **number_of_guests** >= 7
8. Display reservations where **restaurant_name** NOT IN ('Paradise Biryani') AND **is_confirmed** = TRUE
9. Display customers whose **special_request** IS NOT NULL AND **city** IN ('Bangalore', 'Ahmedabad')
10. Display all reservations ordered by **reservation_date** DESC, **reservation_time** ASC
11. Display the **latest 4 reservations** based on **booking_datetime**
12. Display **customer_name**, **restaurant_name**, **reservation_date** where **number_of_guests** > 4 AND **reservation_date** NOT IN ('2024-04-15', '2024-05-10') ordered by **reservation_date** DESC

10. PRACTICAL QUESTION – EVENT TICKET BOOKING DATABASE

You are developing an **Event Ticket Booking System** for managing events across multiple cities.

A. Table Creation

Create a table named **Event_Bookings** with the following attributes:

Column Name	Description	Data Type
booking_id	Unique booking ID	INT
user_name	Name of the user	VARCHAR(100)
event_name	Name of the event	VARCHAR(100)
event_city	City where event is held	VARCHAR(40)
event_date	Date of the event	DATE
event_time	Time of the event	TIME
ticket_count	Number of tickets booked	INT
ticket_price	Price per ticket	DECIMAL(7,2)
is_paid	Payment status	BOOLEAN
booking_datetime	Date & time of booking	DATETIME

B. Alter Table

Modify the table by:

- Adding a new column named **seat_type**
- It represents the category of seating
- Data type should be VARCHAR(30)

C. Rename Table

Rename **Event_Bookings** to **Event_Ticket_Records**.

D. Insert Data

Insert at least 10 records into **Event_Ticket_Records** using the sample data below:

booking_id	user_name	event_name	event_city	event_date	event_time	ticket_count	ticket_price	is_paid	booking_datetime	seat_type
1	Rohit Sharma	Music Fest	Bangalore	2024-06-20	18:30:00	2	2500.00	TRUE	2024-06-10 10:00:00	Gold
2	Anjali Mehta	Tech Summit	Hyderabad	2024-07-05	09:00:00	1	3500.00	TRUE	2024-06-25 14:20:00	Silver
3	Karthik R	Comedy Show	Chennai	2024-06-28	20:00:00	4	1500.00	FALSE	2024-06-26 11:15:00	Regular
4	Pooja Singh	Startup Expo	Delhi	2024-05-15	10:00:00	3	2000.00	TRUE	2024-05-01 09:30:00	Silver
5	Arjun Verma	Music Fest	Mumbai	2024-07-10	19:00:00	5	2800.00	TRUE	2024-07-01 16:45:00	Gold

bookin g_id	user_nam e	event_na me	event_city	event_date	event_ti me	ticket_ count	ticket_pric e	is_paid	booking_dat etime	seat_typ e
6	Sneha Roy	Art Workshop	Kolkata	2024-06-12	15:00:00	2	1200.00	FALSE	2024-06-10 08:50:00	Regular
7	Aman Gupta	Tech Summit	Pune	2024-07-05	09:00:00	1	3500.00	TRUE	2024-06-29 17:00:00	Platinum
8	Neha Jain	Comedy Show	Jaipur	2024-06-22	21:00:00	3	1600.00	TRUE	2024-06-20 13:40:00	Regular
9	Ramesh N	Startup Expo	Bangalore	2024-04-18	10:30:00	2	1800.00	FALSE	2024-04-15 12:00:00	Silver
10	Kavya Patel	Art Workshop	Ahmedabad	2024-07-15	14:00:00	4	1300.00	TRUE	2024-07-05 09:10:00	Gold

E. Update Data

Update **is_paid** to **TRUE**

for bookings where:

- **event_date** is in **July 2024**
- **AND ticket_count** is greater than **3**

F. Delete Data

Delete booking records where:

- **is_paid** = **FALSE**
- **AND event_date** is before **2024-05-01**

QUERY OPERATIONS

1. Display bookings where **event_date** **BETWEEN '2024-06-01' AND '2024-06-30'** **AND ticket_count** **NOT BETWEEN 2 AND 4**
2. Display records where **event_name** **IN ('Music Fest', 'Comedy Show')** **AND is_paid** = **TRUE**
3. Display users from **Bangalore OR Hyderabad** **AND NOT** attending events in **April 2024**
4. Display bookings where **seat_type** **IS NULL OR ticket_price** > **3000**
5. Display bookings where **event_city** **NOT IN ('Delhi', 'Mumbai')** **AND ticket_count** >= **3**
6. Display users who booked **BETWEEN '2024-06-10' AND '2024-07-01'** **AND booking_datetime** **NOT BETWEEN '2024-06-10 00:00:00' AND '2024-06-20 23:59:59'**
7. Display bookings where **event_time** **BETWEEN '18:00:00' AND '21:00:00'** **AND is_paid** = **TRUE**
8. Display records where **seat_type** **IN ('Gold', 'Platinum')** **AND ticket_price** **NOT BETWEEN 2000 AND 3000**
9. Display bookings where **event_name** **NOT IN ('Art Workshop')** **AND ticket_count** > **2**
10. Display users whose **booking_datetime** **IS NULL OR is_paid** = **FALSE**

11. Display bookings from **Pune OR Ahmedabad AND seat_type = 'Gold'**
12. Display bookings where **event_date = '2024-07-05' OR ticket_price BETWEEN 1500 AND 1800**
13. Display bookings where **is_paid = TRUE AND event_city NOT IN ('Jaipur', 'Kolkata')**
14. Display all bookings ordered by **event_date DESC, event_time ASC**
15. Display **user_name, event_name, ticket_count**
where **ticket_price > 2000 AND event_date NOT IN ('2024-04-18', '2024-05-15')** ordered by **ticket_price DESC**